
Mini Project Report

PROJECT TITLE: CRIME SCENE NETWORK SIMULATOR

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DEPARTMENT: DEPARTMENT OF COMPUTING AND CYBER SECURITY

1. Abstract

The Crime Scene Network Evidence Simulator is a mini project developed using Cisco Packet Tracer to simulate a real-world enterprise network and demonstrate the fundamentals of digital forensic investigation. The project consists of multiple departments connected through Virtual Local Area Networks (VLANs), including the Investigation Department, Evidence Storage Department, and Administration Department. An Evidence Server is configured with DNS and HTTP services to simulate secure storage and access to digital evidence.

Network traffic such as ICMP (Ping), DNS, and HTTP is generated to simulate normal user activities and suspicious access attempts. Packet Tracer's Simulation Mode is used to observe packet flow and analyze network communication between devices. Security mechanisms such as Access Control Lists (ACLs), Port Security, SSH, DHCP, Syslog and FTP are implemented to protect the network and prevent unauthorized access.

The project demonstrates the integration of networking, cybersecurity, and digital forensics by creating a simulated cybercrime investigation environment. It provides practical knowledge of network design, traffic analysis, evidence collection, and network security. The expected outcome is to understand how digital evidence can be generated, analysed, and protected in a secure enterprise network.

2. Objective

The main objective of this project is to design and implement a secure enterprise network that simulates a digital crime scene using Cisco Packet Tracer. The project aims to configure VLANs, inter-VLAN routing, DNS and HTTP services, generate network traffic, analyse communication, and implement security features such as ACLs to demonstrate the basic concepts of network forensics and digital evidence analysis.

3. Problem Statement

Organizations store sensitive digital information on their networks, making them potential targets for unauthorized access and cyber-attacks. Without proper network segmentation and security mechanisms, confidential data can be compromised. This project addresses the need for a secure network environment by simulating a digital crime scene where suspicious

network activity can be identified, analysed, and controlled using networking and digital forensic techniques.

4. Tools & Software

- Cisco Packet Tracer
- Windows 11
- Command Prompt (CMD)
- Cisco IOS CLI
- DNS Service
- HTTP Service
- FTP Service
- DHCP Service

5. Methodology

The project was implemented using the following steps:

1. Design the enterprise network topology.
2. Add routers, switches, servers, and client PCs.
3. Create VLANs for different departments.
4. Configure switch ports and trunk links.
5. Configure Router-on-a-Stick for inter-VLAN routing.

6. Assign IP addresses to all devices.
7. Configure DNS and HTTP services on the Evidence Server.
8. Generate ICMP, DNS, and HTTP traffic.
9. Analyse packet flow using Packet Tracer Simulation Mode.
10. Configure ACLs and other security features.
11. Test the network and verify the results.

Flow:

Start → Design Network → Configure VLANs → Configure Router → Assign IP Addresses → Configure Servers → Generate Traffic → analyse Packets → Implement Security → Test Network → End

6. Implementation

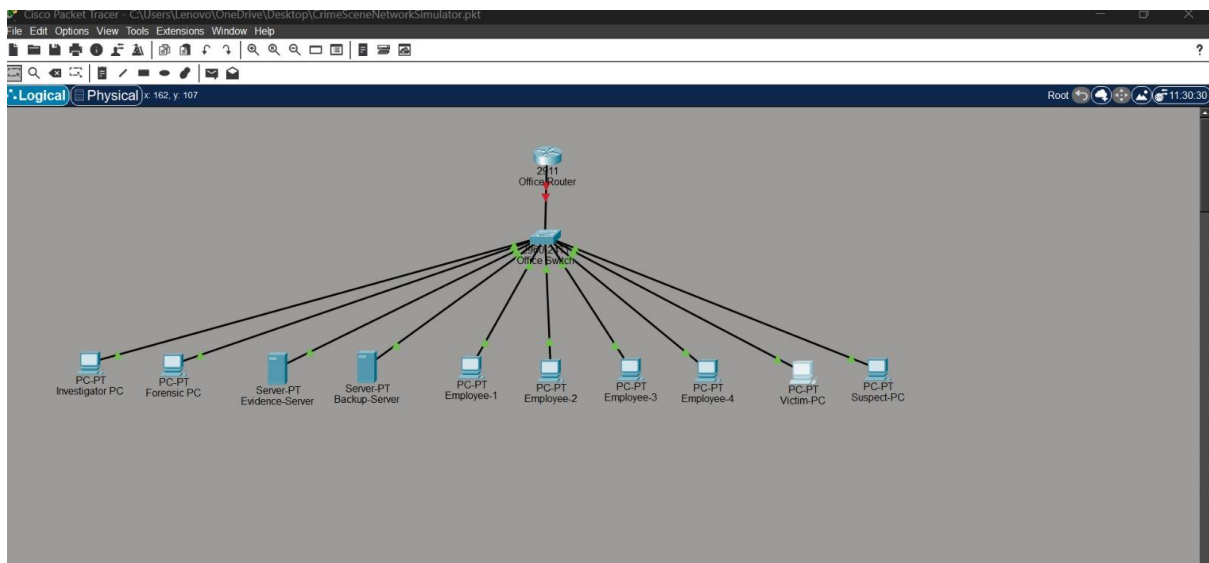
The implementation includes:

- Network topology creation
- VLAN configuration
- Router-on-a-Stick configuration
- IP address configuration
- DNS server configuration
- HTTP server configuration
- Traffic generation using Ping, DNS, and HTTP

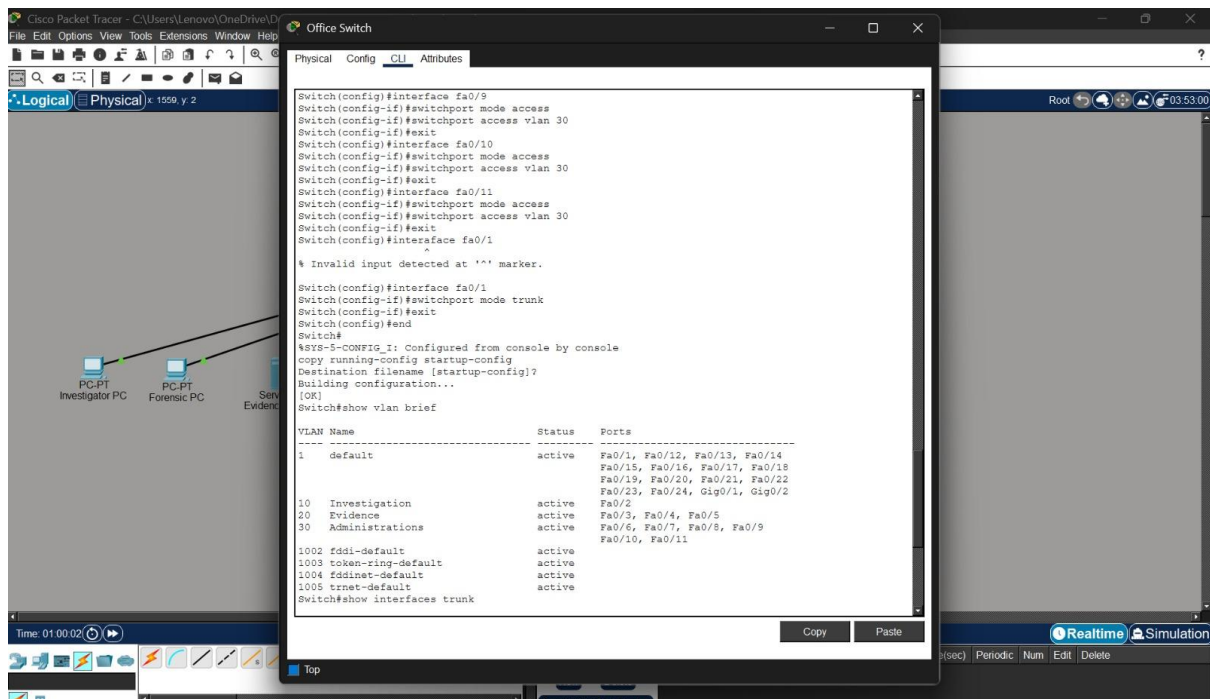
- Packet analysis using Simulation Mode
- ACL configuration
- Port Security configuration
- SSH configuration
- DHCP configuration
- FTP configuration

Proofs:

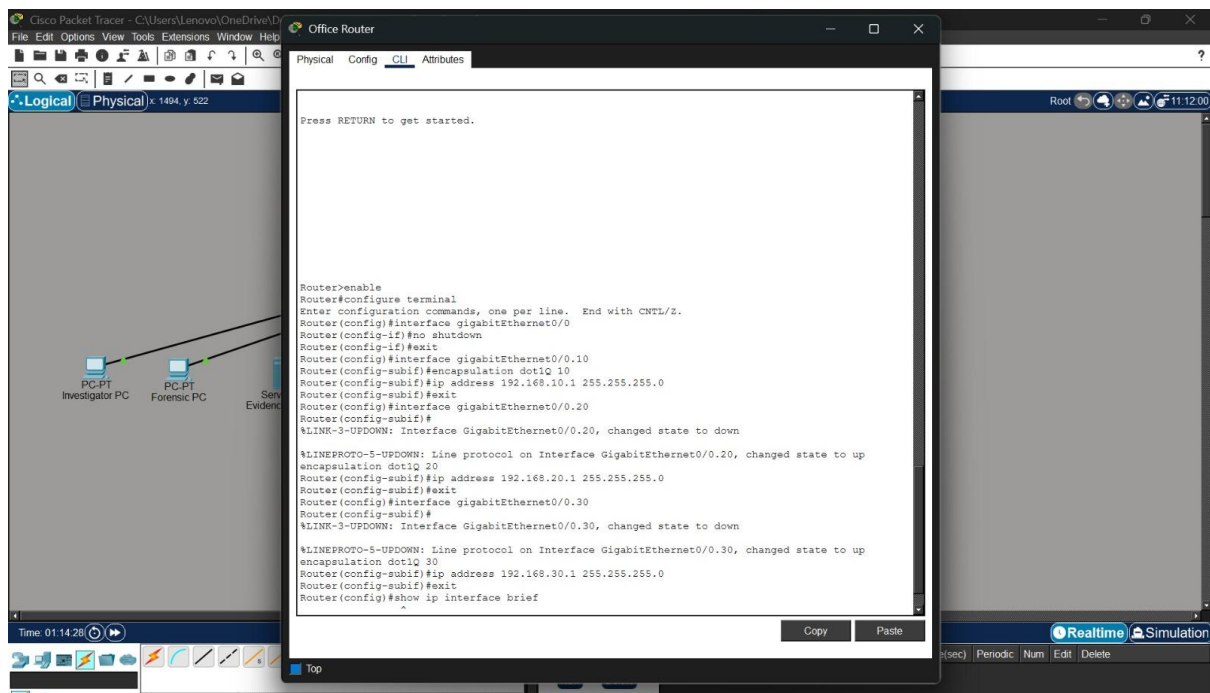
Network Topology basic setup:



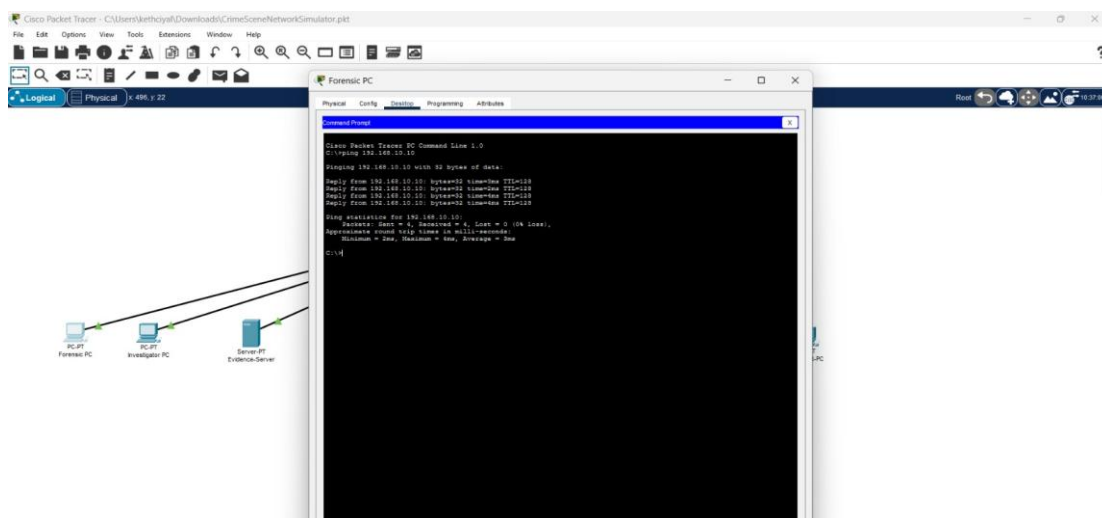
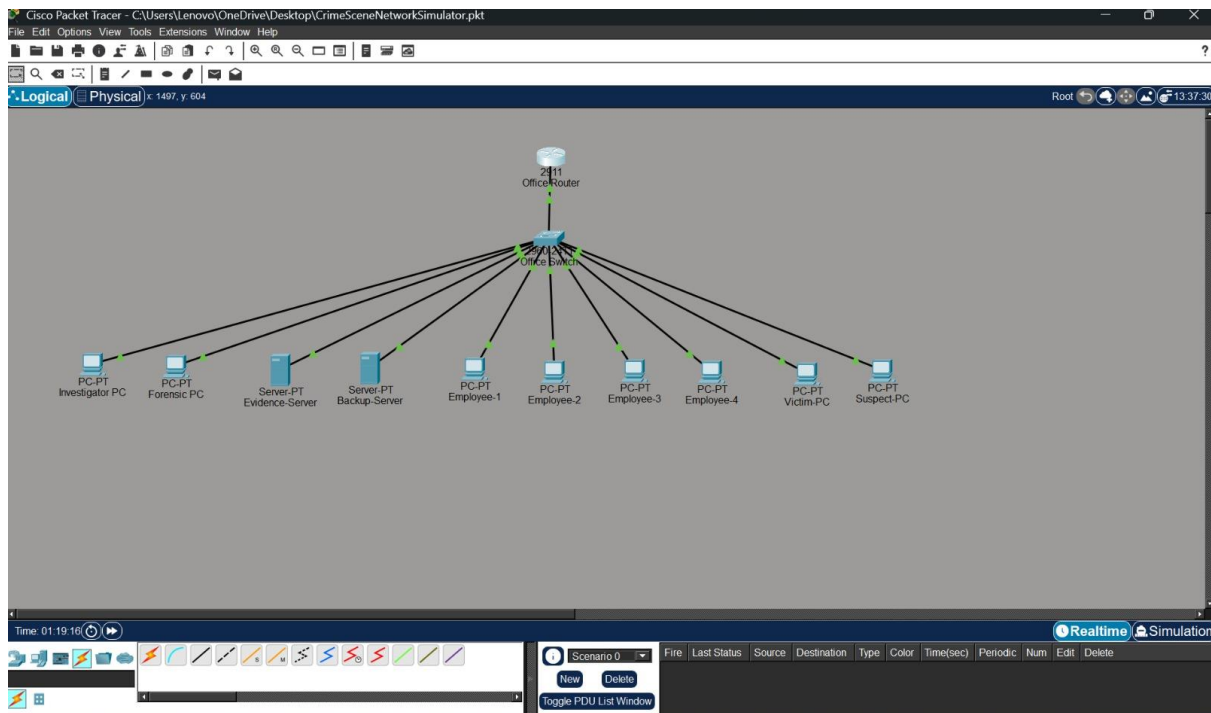
Switch Configuration:



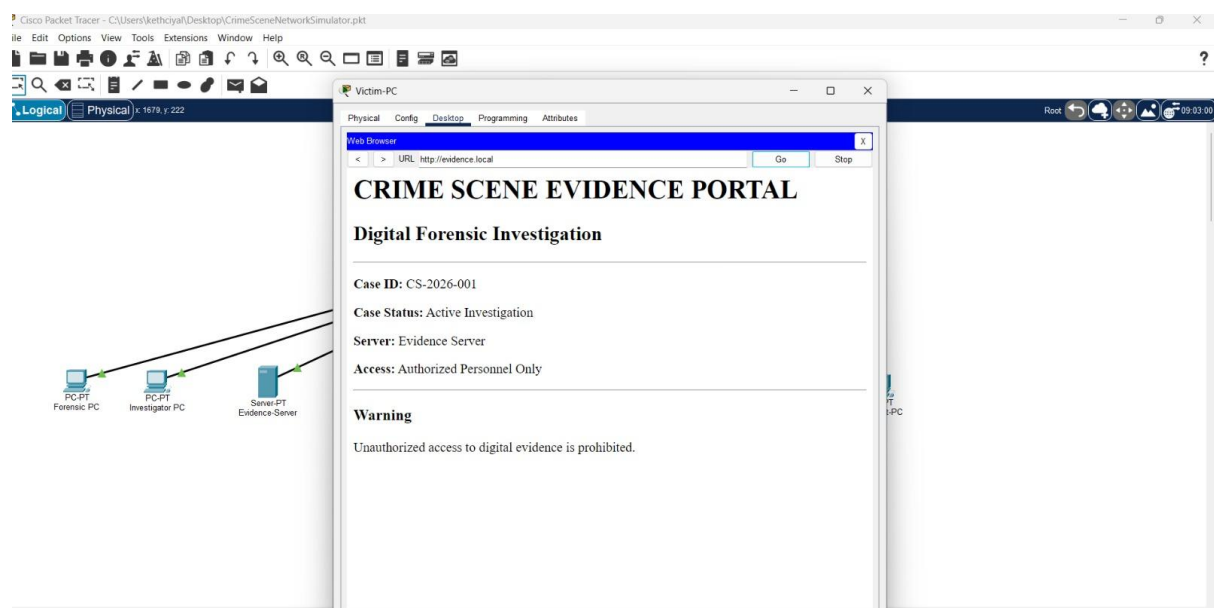
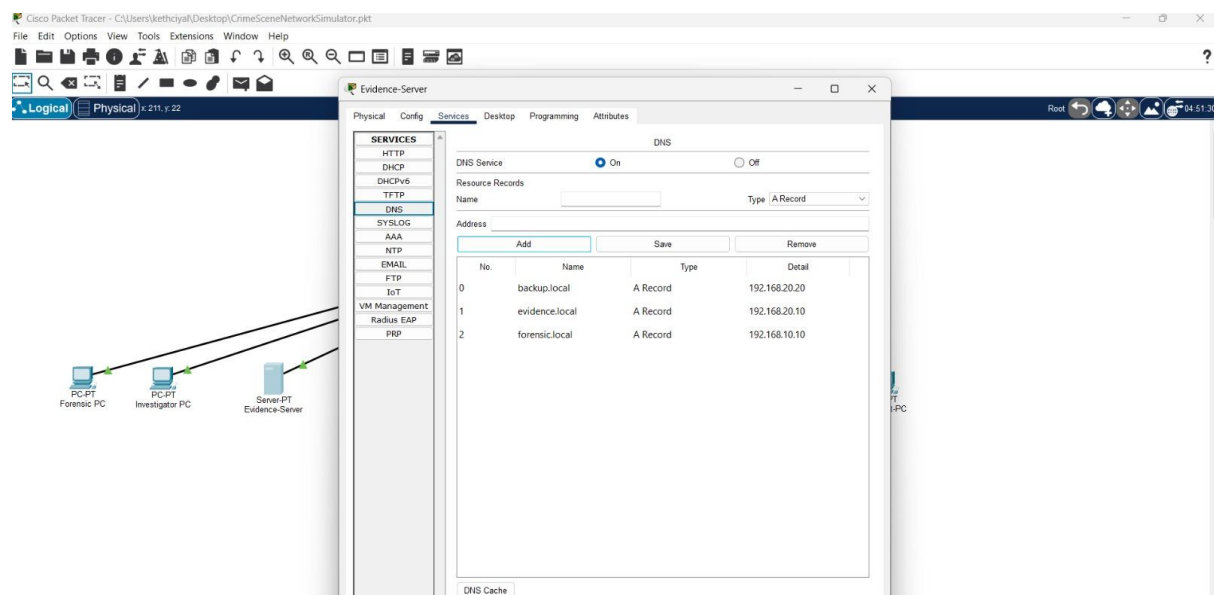
Router Configuration:



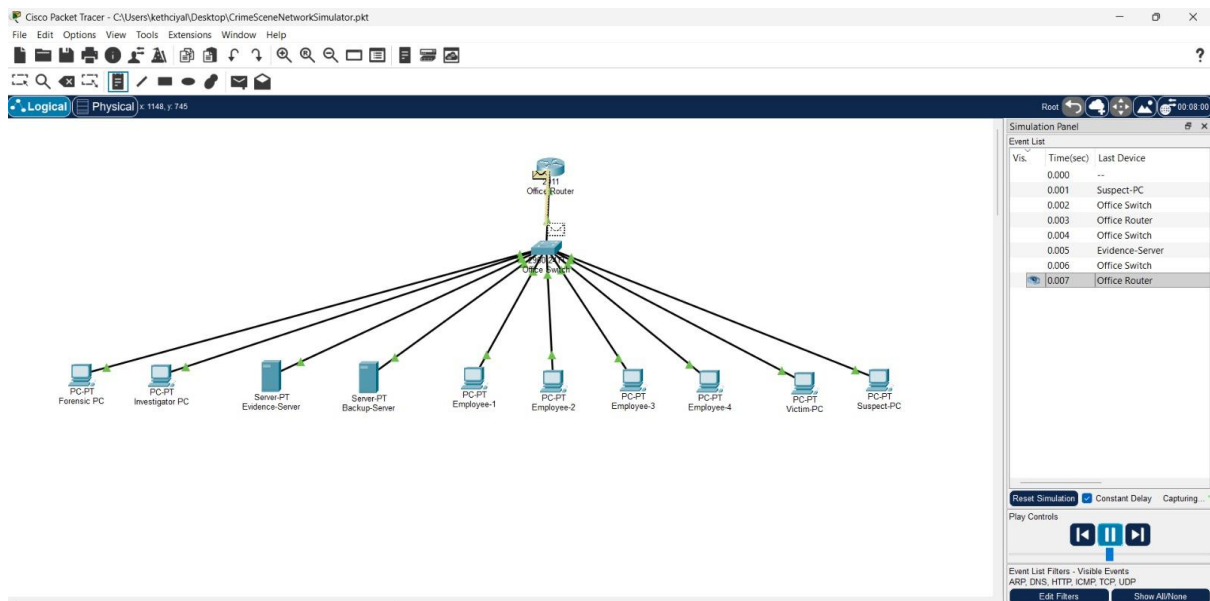
Configure All PC's & Servers: (including ping)



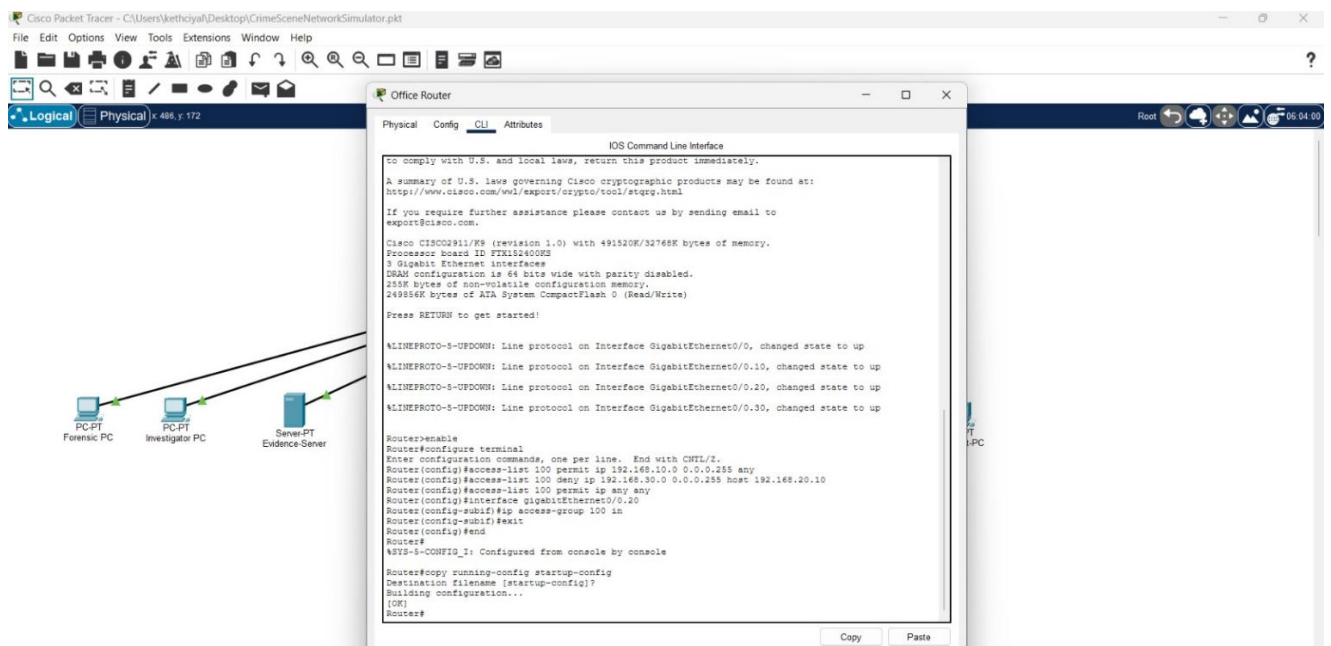
Configure the Evidence Server (DNS+HTTP):



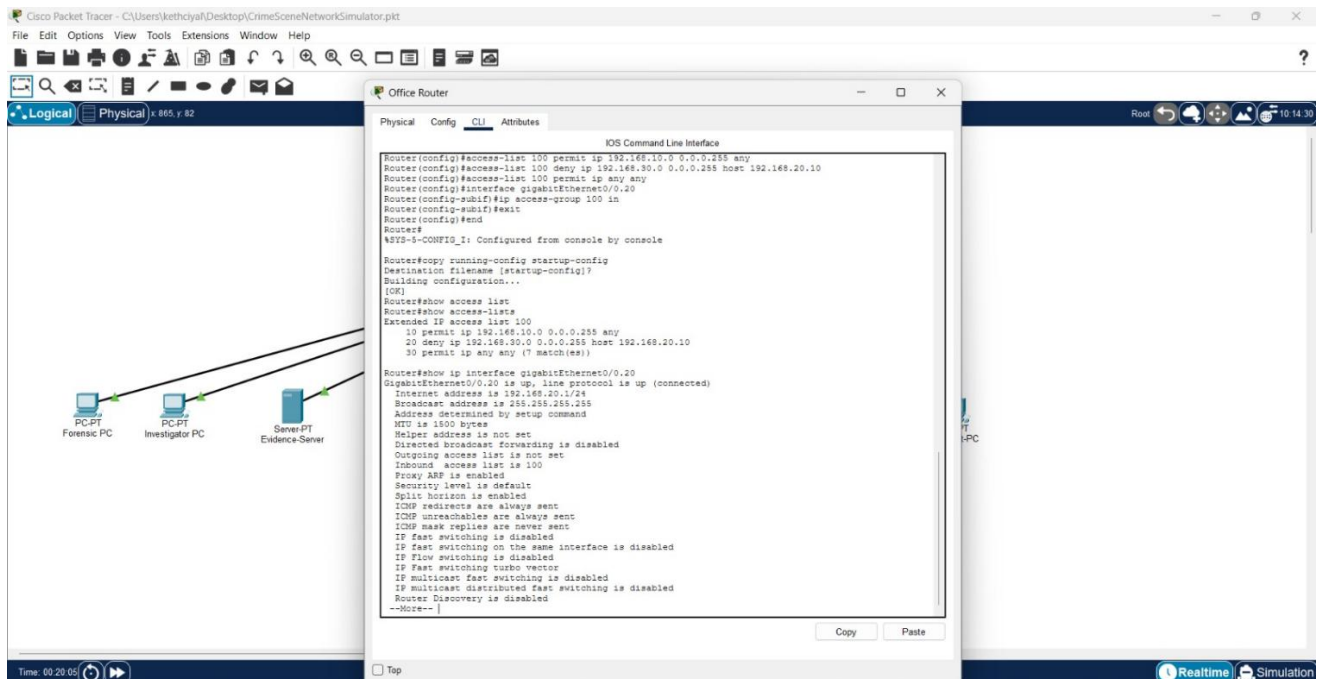
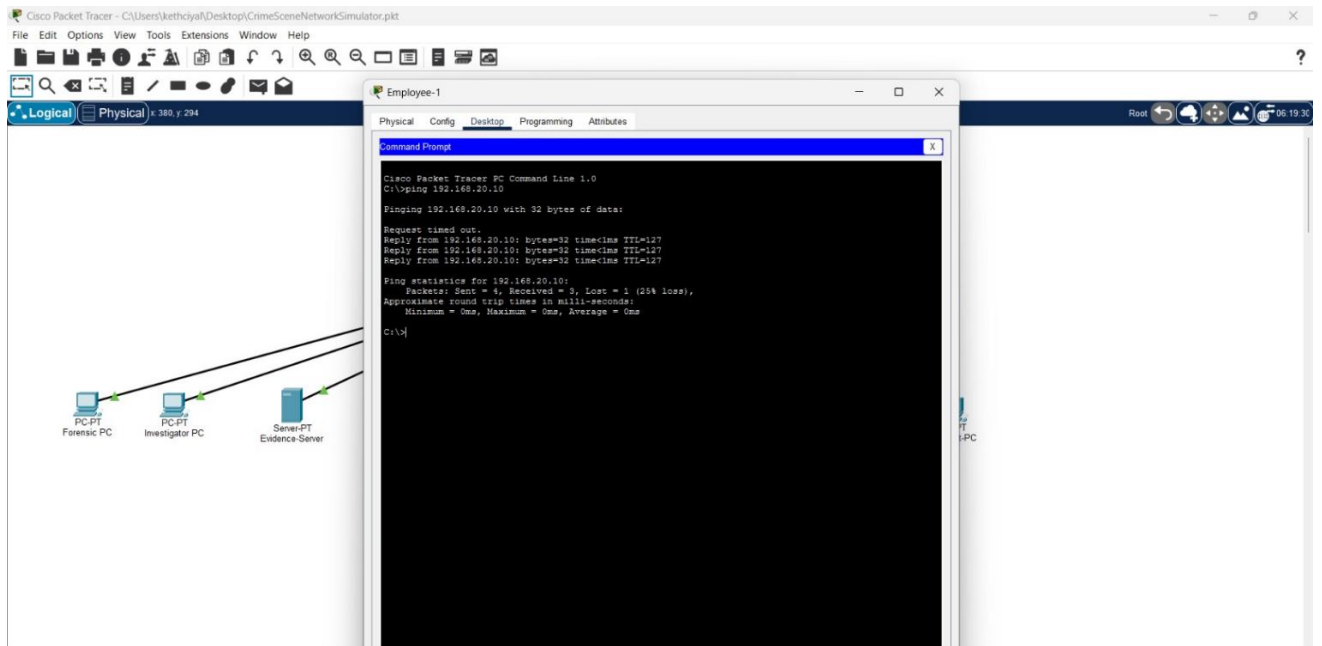
Simulation:

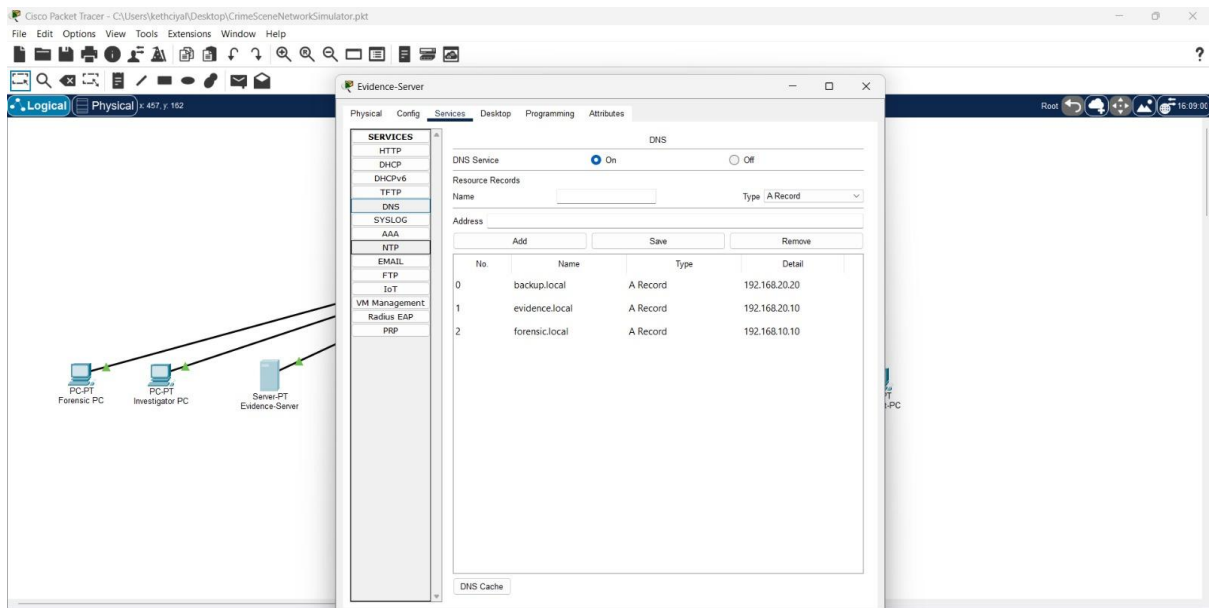


Secure the Evidence Server Using ACL (Access Control List):

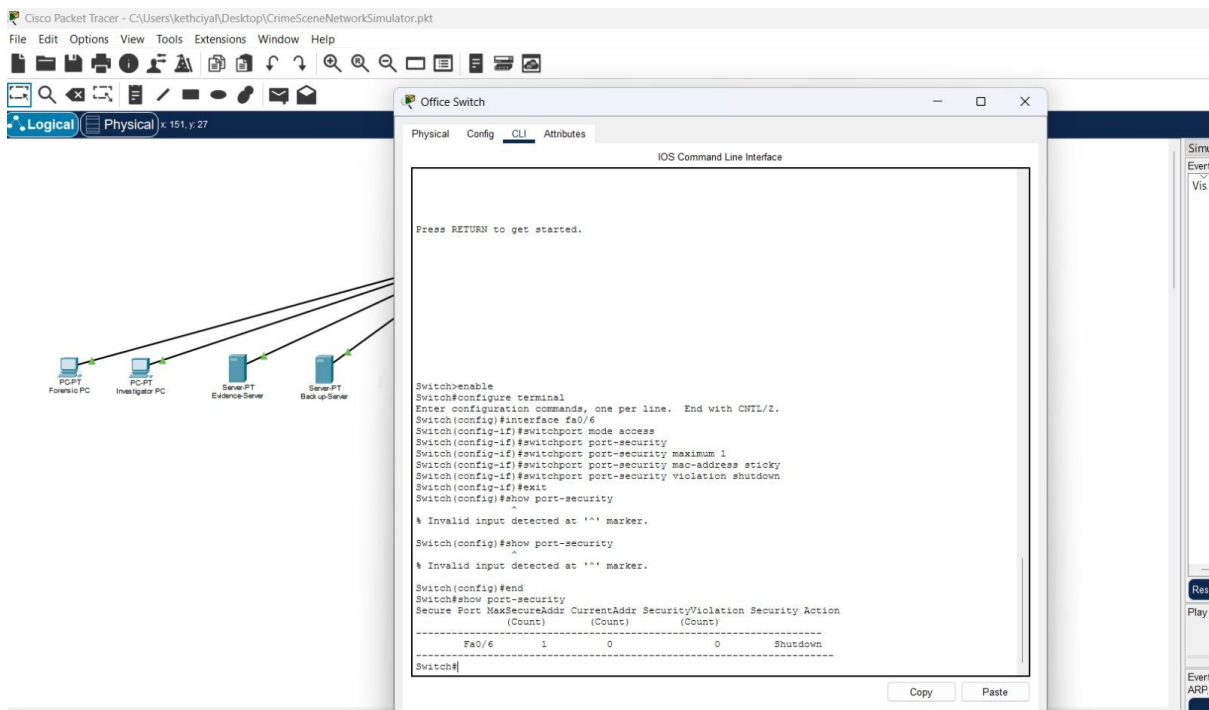


Testing ACL:

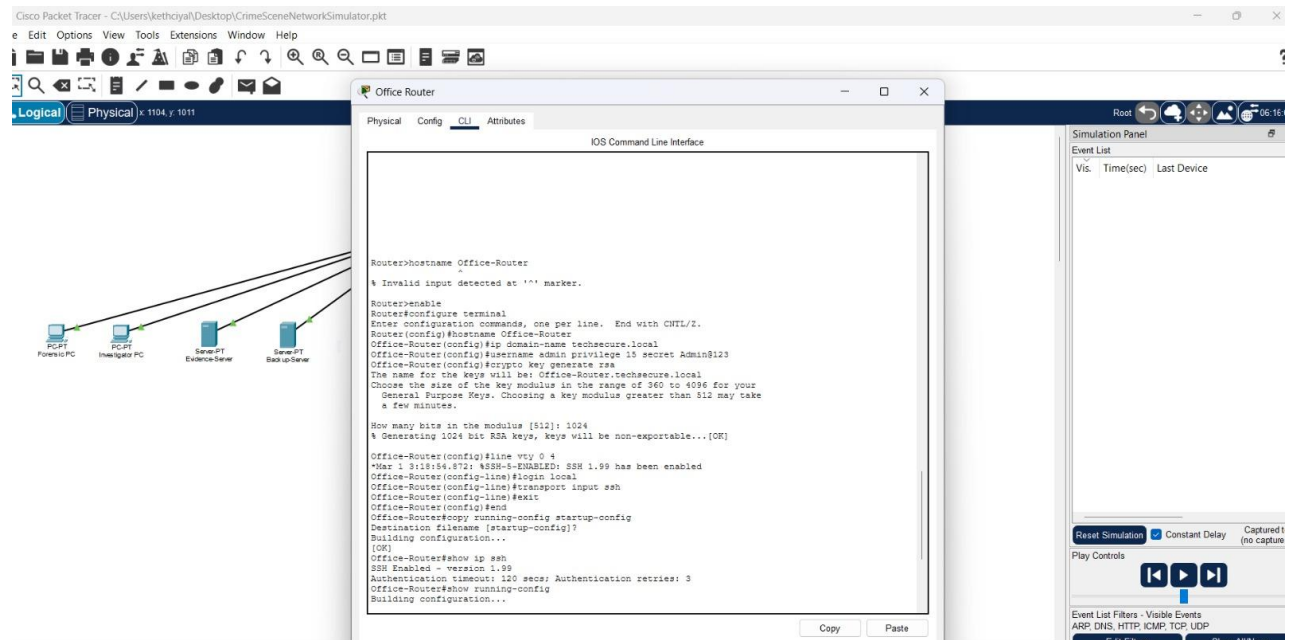




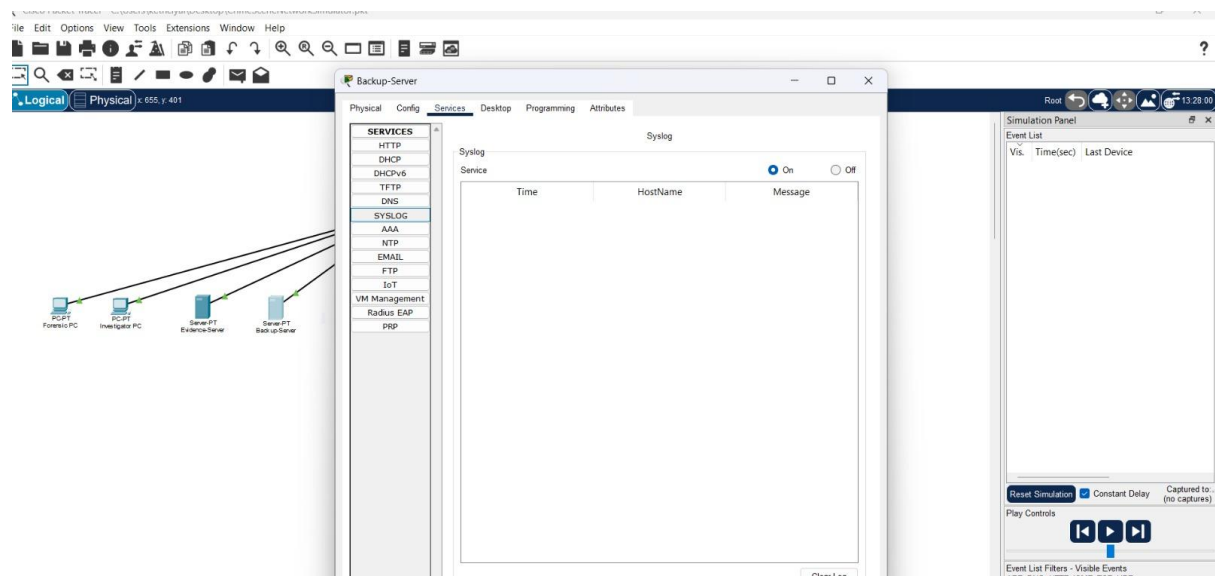
Feature 1 :Configure Port security



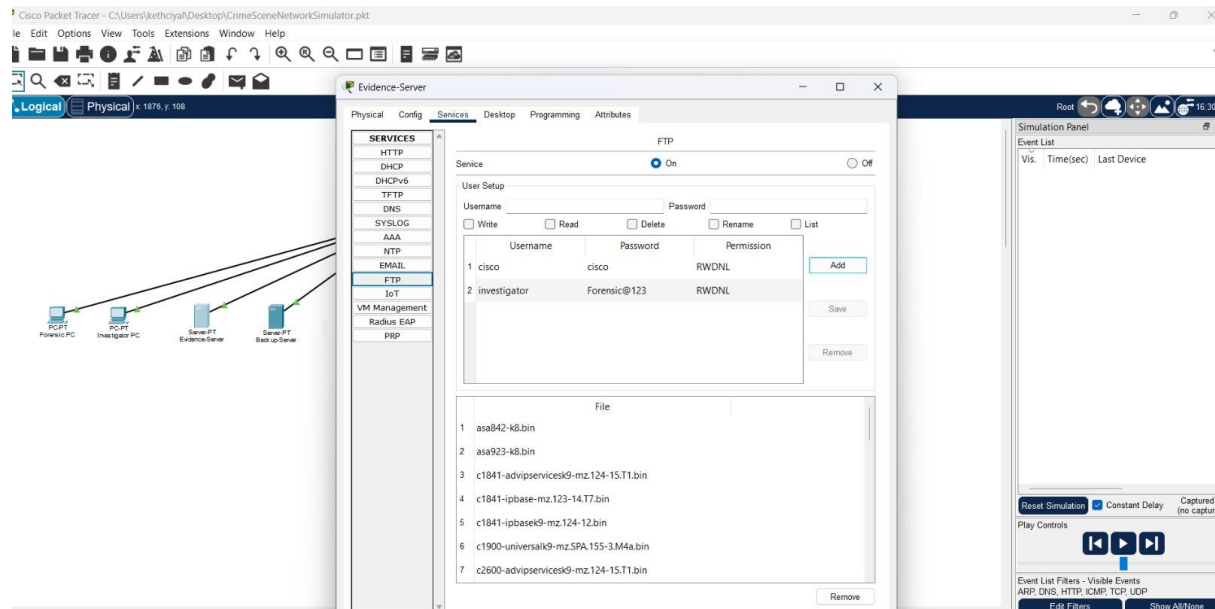
Feature 2 : Configure Secure Remote Management (SSH) :



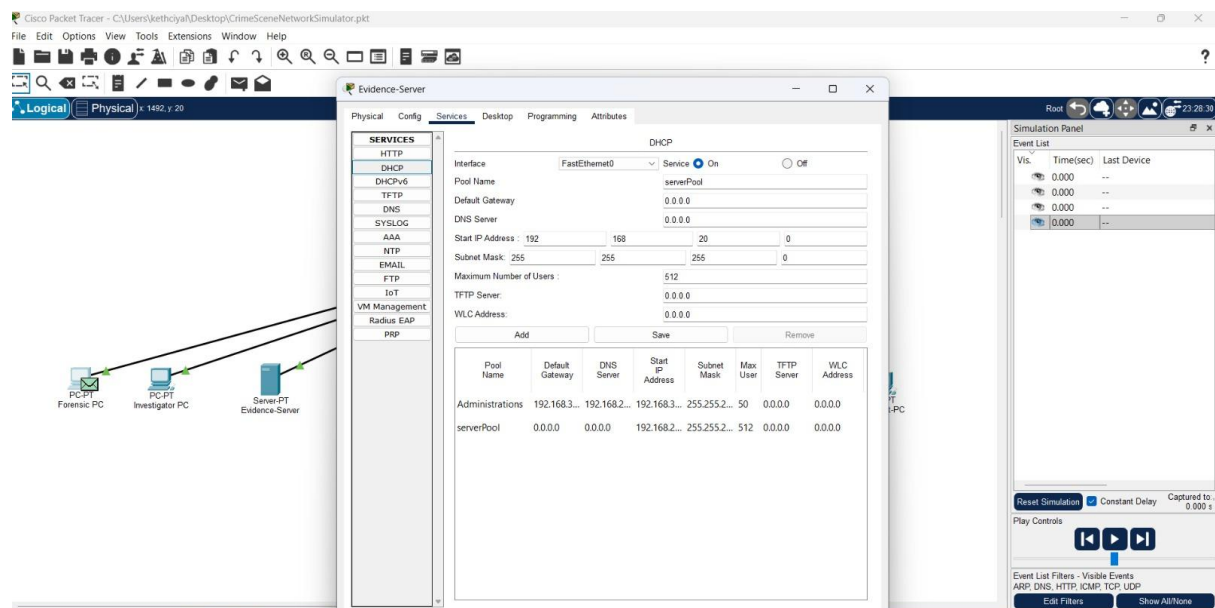
Feature 3 – Configure Syslog Server :



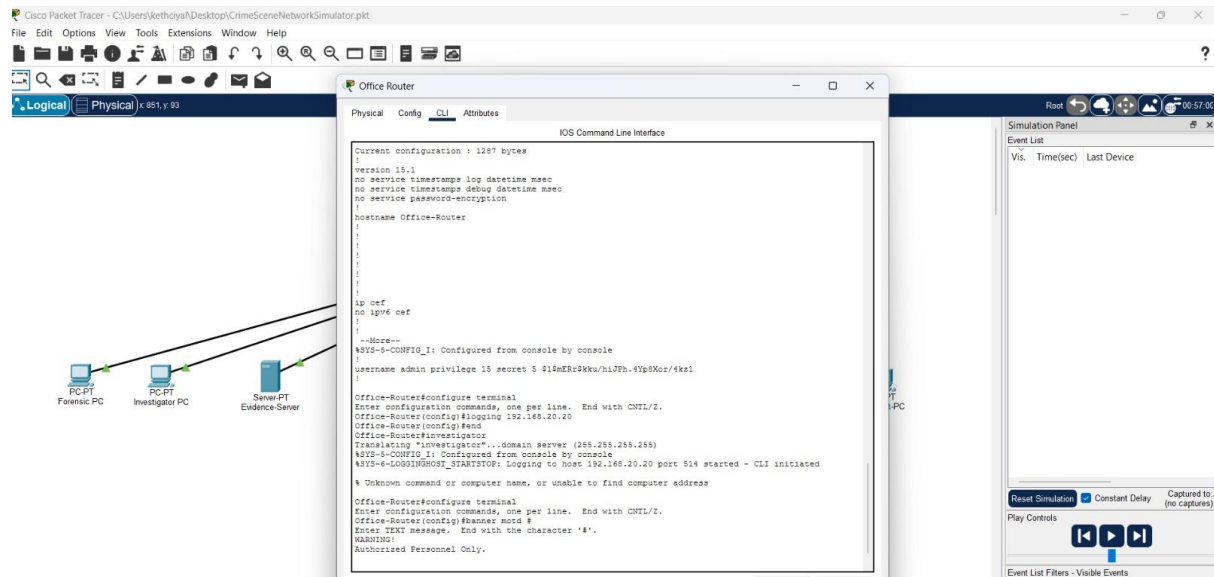
Feature 4 : Configure FTP server :



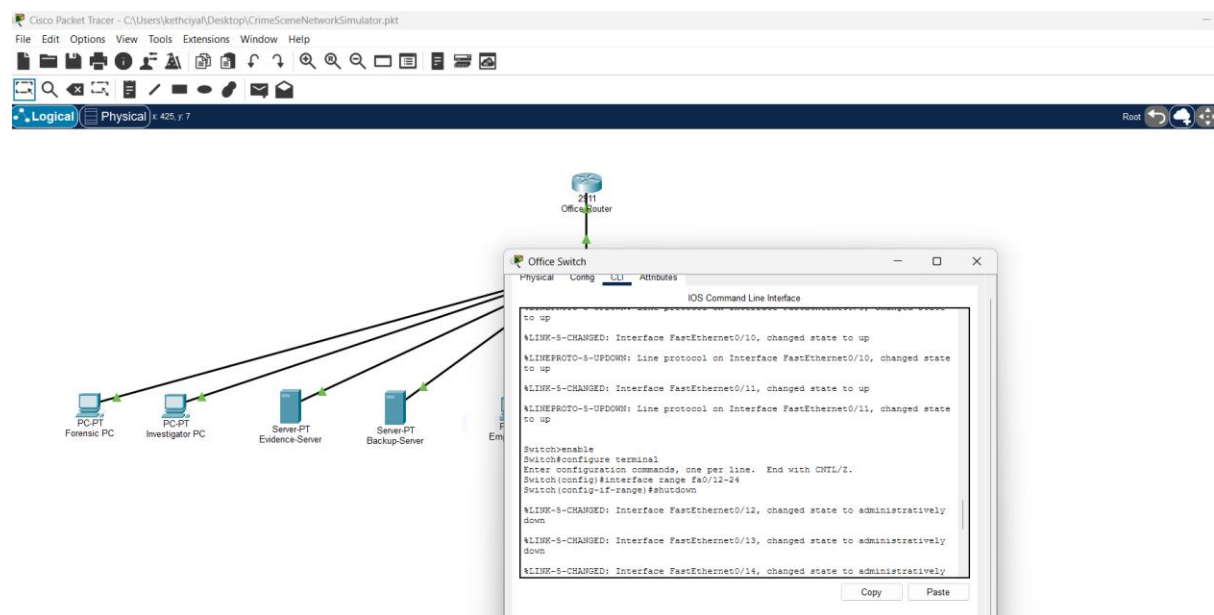
Feature 5: Configure DHCP



Feature 6 : Banner Message (Warn Unauthorized Users)



Feature 7 : Disable Unused Switch Ports



7. Results

The network was successfully configured and all devices communicated as expected. VLANs isolated different departments, while Router-on-a-Stick enabled communication between VLANs. DNS and HTTP services functioned correctly, allowing users to access the Evidence Server. Network traffic was successfully generated and analysed using Packet Tracer's Simulation Mode. The ACL effectively restricted unauthorized access to the Evidence Server, demonstrating improved network security and basic digital forensic investigation.

8. Real-Time Incident Scenario and Project Relevance

In an organization, employees, servers, and other systems are connected through a computer network. If an attacker or an unauthorized employee gains access to the internal network, they may attempt to discover important systems, access confidential information, or communicate with sensitive servers.

During such an incident, investigators need to determine:

Which system generated the suspicious activity?

What IP address was involved?

Which server was contacted?

Which protocols were used?

When did the communication occur?

Was the access authorized or unauthorized?

How can the sensitive server be protected from further access?

The Crime Scene Network Evidence Simulator demonstrates these investigation steps in a controlled environment.

Simulated Real-Time Incident:

The project uses a fictional organization called TechSecure Pvt. Ltd.

The organization contains three major network segments:

VLAN	Department	Purpose
VLAN10	Investigations	Forensic Investigations
VLAN20	Evidence Storage	Stores Sensitive digital evidence
VLAN30	Administration	Employees and regular users

The Evidence Server is placed in VLAN 20 so that sensitive information is separated from normal employee systems.

A Victim PC and a Suspect PC are included in the Administration network. The Suspect PC is used to simulate a workstation involved in suspicious communication with the Evidence Server.

This project helps to:

- Identify suspicious network communication.

- Find the source and destination IP addresses.
- Understand network packet flow.
- Analyze basic network evidence.
- Protect sensitive servers using ACLs.
- Understand how network security supports digital forensics.

Real-World Applications

This concept can be useful in:

- Cybercrime investigations
- Corporate security
- SOC monitoring
- Incident response
- Digital forensic investigations

9. Conclusion

The Crime Scene Network Evidence Simulator successfully demonstrated the integration of networking, cybersecurity, and digital forensics in a simulated enterprise environment. The project provided practical experience in configuring VLANs, routing, network services, and security mechanisms while analysing network traffic for forensic investigation. It improved understanding of digital evidence collection, packet analysis, and network protection, making it an effective learning project for network security and digital forensics.